

Loading-Controlled Stiffening in Nanoconfined Ionic Liquids

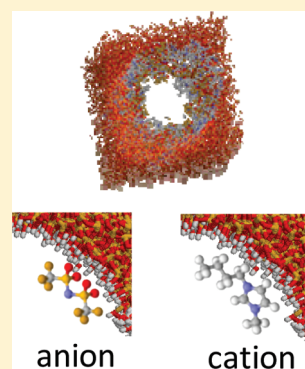
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S Supporting Information

ABSTRACT: An important strategy for using ionic liquids is to immobilize them by impregnation of supports or incorporation into porous solids to obtain materials called “ionogels”. Of considerable importance for applications (electrolyte membranes, supported catalysts, etc.), such confinement results in dramatic changes in the physicochemical properties of the ionic liquid. Here, we report molecular simulations of a silica nanopore that is gradually filled with a typical imidazolium salt ionic liquid to obtain a realistic model of these ionogels. Despite the significant layering and stiffening of the ionic liquid in the vicinity of the silica surface, the pair correlation functions and magnitude of its dynamics clearly evidence liquid-like behavior. An increase in the self-diffusivity and ionic conductivity, associated with a decrease in the characteristic residence times of ions at the silica surface, is observed upon increasing the loading as the ionic liquid fills the nanopore center and tends to recover its bulk properties.

SECTION: Surfaces, Interfaces, Catalysis



Ionic liquids (ILs) are “liquids composed solely of ions”. Usually, the term is restricted to salts having melting points below 100 °C (higher-temperature melts are referred to as molten salts). Their properties can be tuned by the cation/anion combination. Features such as negligible vapor pressure, non-flammability, high ionic conductivity, and a wide potential window make ILs the subject of research as alternative solvents in organic synthesis and organometallic catalysis,¹ biocatalysis,² and electrochemistry.³ There is a practical advantage in immobilizing ILs by impregnation of a support, the IL being used as an adsorbed film, or by incorporation into a porous substrate, typically as solid-like systems called ionogels.^{4,5} This approach is well illustrated by the development of supported ionic liquid catalysis based on immobilizing metal catalysts within IL films supported on silica beads.^{6,7} Several applications of silica ionogels have been reported in various fields such as solid electrolytes,⁸ sensing,⁹ and drug release.¹⁰

Properties of ILs are modified upon confinement; these include glass transition, crystallization and melting temperatures, fluorescence and vibrational spectra,¹¹ and ionic conductivity.¹² There is, to date, little information on the interfacial structure of ILs in contact with a solid surface. Moreover, this information appears contradictory, giving rise to controversial issues; solid-like ordering has been observed in IL films on silica,^{13,14} whereas a liquid behavior has been evidenced by NMR in ionogels.¹⁵ To gain insights into the behavior of confined ILs, we report molecular dynamics simulations of a realistic model of a silica nanopore that is gradually filled with a typical imidazolium salt ionic liquid (see Supporting Information for the computational details). The cylindrical pore of diameter $D = 4.8$ nm is described at the atomistic level to address in a realistic way the effect of confinement and surface forces on the behavior of the IL (Figure 1). After carving the pore out of amorphous silica, its

electroneutrality is ensured by saturating all O dangling bonds with H atoms. The IL, which corresponds to a typical IL in ionogels, is composed of bis[(trifluoromethyl)sulfonyl]amide anions and 1-butyl-3-methylimidazolium cations (Figure 1). The electrostatic potential created by the anion and cation in their trans and cis conformations was estimated using ab initio DFT calculations (see the Supporting Information). For both conformations, a negative electrostatic potential is observed around the anion with the N, F, and O atoms being negatively charged and the C and S atoms being positively charged. In contrast, both conformations of the cation, which exhibit a more homogeneous positive potential, depart from traditional H-donors that exhibit negatively charged regions.¹⁶ Different impregnation loadings were modeled where a fraction of $f = 0.25, 0.50$, or 1 of the porosity was filled.

The density of the IL in the pore is shown in Figure 2. For all loadings, density oscillations are observed for both the anions and cations, which reveal the layering of the IL at the surface. Such layering, which is characteristic of fluids in pores or on surfaces, has been already reported for ILs on sapphire,¹⁷ gold,¹⁸ and mica.^{19,20} For $f = 0.25$, the pore accommodates a layer of cations and anions adsorbed at the surface while the IL is depleted in the pore center. For $f = 0.50$, a cation–anion pair layer forms on the first layer, and the IL occupancy in the pore center becomes non-negligible. For $f = 1$, the pore accommodates four cation layers and three anion layers; beyond the layer of cations/anions in contact with the surface, the cations form layers at well-defined distances, and the anions form layers between the cation layers. The formation of this “sandwiched”

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structure beyond the first layer is related to the particular pore size, which favors this arrangement with respect to a structure of

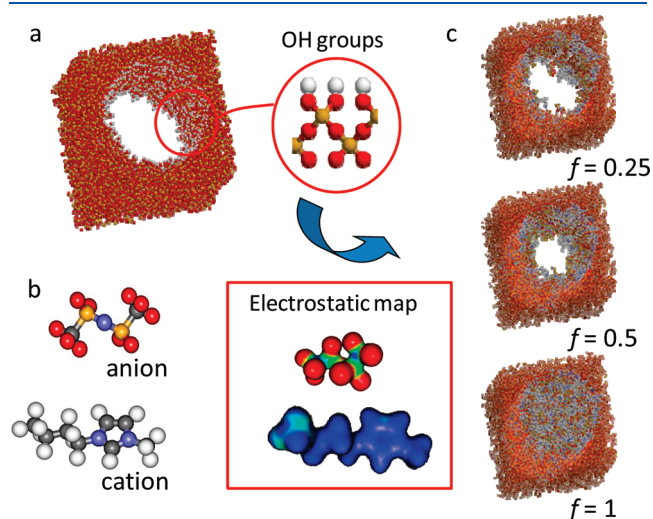


Figure 1. (a) Atomistic silica nanopore of diameter $D = 4.8$ nm. Orange and red spheres are silicon and oxygen atoms, respectively. White spheres are hydrogen atoms of the OH groups that delimit the pore surface. The box size is $6.4 \times 6.4 \times 9.3$ nm³. (b) Imidazolium-based IL with bis[(trifluoromethyl)sulfonyl]amide anions (NTf₂⁻) and 1-butyl-3-methylimidazolium cations (C₄mim⁺). For both ions, an electrostatic potential mapped onto the isodensity surface with 0.03 e/bohr³ is shown. The electrostatic potential scale, which is linear from negative to positive values, increases from red, to orange, to yellow, to green, to blue. (c) Typical molecular configurations showing the silica nanopore shown in (a) when a fraction of $f = 0.25$, 0.5, and 1 of the porosity is filled with the IL.

n layers of cation/anion pairs. In agreement with simulations for structureless pores,²¹ density oscillations remain marked at large distances from the surface despite the atomic surface corrugation of our model. Despite the ordering of the IL at the surface, the pair correlation functions $g(r)$ in Figure 3 show that it retains a liquid-like structure as only short-range order is observed. Moreover, the correlation length ξ , which is obtained from the exponential decay between successive maxima in the $g(r)$ function, is nearly identical for the bulk and confined IL, $\xi \approx 1$ nm.

We calculated the self-diffusivity of the anions (D_{anion}) and cations (D_{cation}) and the ionic conductivity σ (see the Supporting Information). D_{cation} , D_{anion} , and σ for the confined IL, which are always lower than the bulk, increase with increasing f (Table 1). In agreement with the work by Neouze et al.,¹² σ varies from $\sim \sigma_0/4$ (σ_0 is the bulk ionic conductivity) at low loading up to $\sim \sigma_0$ when the pore is filled. For $f = 0.25$, D_{cation} , D_{anion} , and σ are very low as most of the adsorbed ions strongly interact with the silica surface. In contrast, at higher loadings, the dynamics of the confined IL becomes faster due to the fact that as further adsorption occurs, the IL fills the pore center and recovers its bulk properties. The small decrease in the conductivity upon decreasing f with respect to that in the self-diffusivity suggests that collective properties (such as conductivity, which is related to the motion of the center of charges) are quantitatively less affected by confinement and surface forces than single properties (such as the self-diffusivity, which is related to the motion of each individual ion). While the conductivity of the bulk IL can be estimated from the ion self-diffusivity using the Nernst–Einstein equation, the conductivity of the confined IL is a subtle combination of a surface contribution arising from ions close to the pore surface and a bulk-like contribution arising from ions in the pore center. Moreover, we note that a drastic slowing down of the

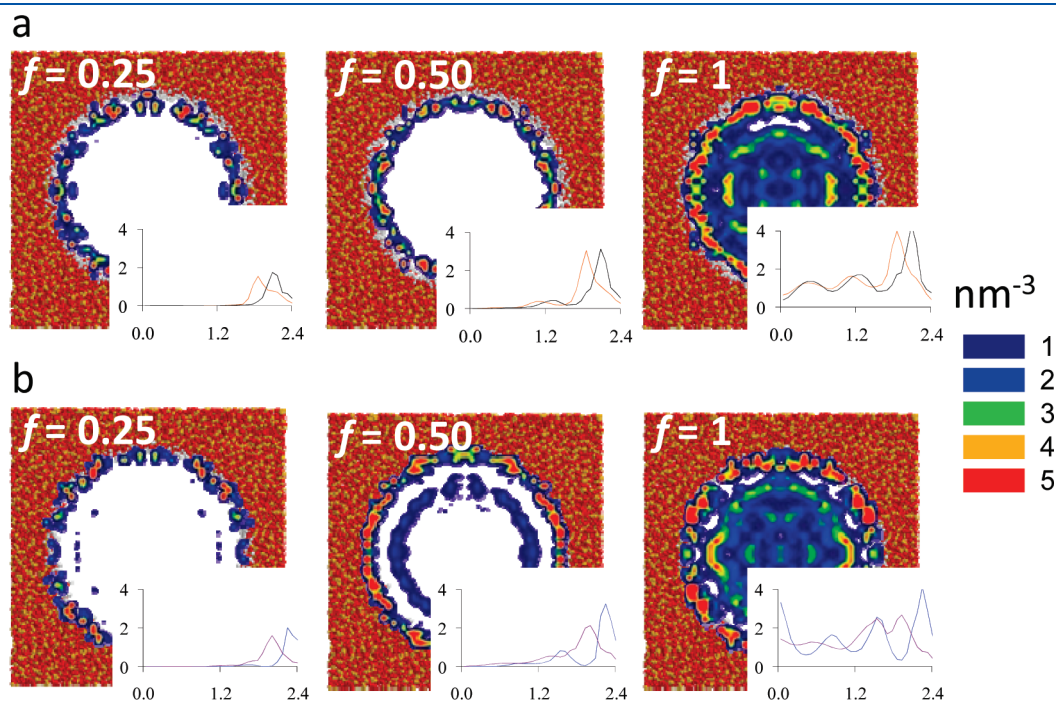


Figure 2. Contour plots showing the density distribution at ambient temperature of (a) anions and (b) cations of an IL at location (x,y) in a silica nanopore. Filling fractions are (from left to right) $f = 0.25$, 0.50, and 1. The scale is linear in local anion density in nm⁻³. The silica pore is also shown for the sake of clarity. Orange and red segments denote silicon and oxygen atoms, while the white segments are hydrogen atoms that delimit the pore surface. For each anion contour plot, the inset shows the radial density profiles in nm⁻³ for the fluorine (orange line) and oxygen (black line) atoms of the anion. For each cation contour plot, the inset shows the radial density profiles in nm⁻³ for the nitrogen atom (blue line) and the last carbon atom of the butyl chain (purple line) of the cation.

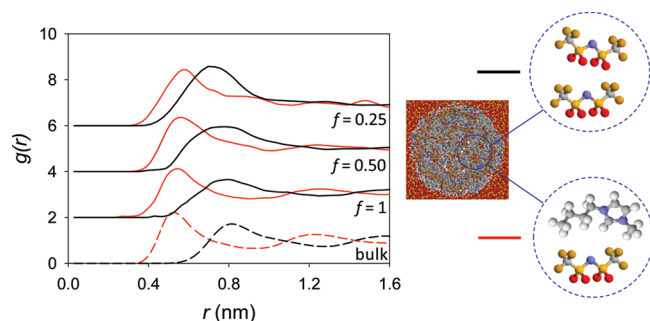


Figure 3. Pair correlation functions $g(r)$ for the IL confined at ambient temperature in the silica nanopore of a diameter $D = 4.8$ nm; the black and red lines are the $g(r)$ functions between the centers of mass of the anions and between the centers of mass of the cations and the anions. The dashed lines are for the bulk, while the solid lines are for the confined ionic liquid when the pore is filled with different loadings: $f = 1$, 0.50, and 0.25. For the sake of clarity, the $g(r)$ functions for the confined IL with $f = 1$, 0.50, and 0.25 have been shifted up by +2, +4, and +6, respectively.

self-diffusivity of the IL does not necessarily imply a drop in the ionic conductivity (for instance, in the Grotthius mechanism, conductivity can be observed without mass transport). A possible interpretation of our results is as follows. For $f = 1$, despite a large decrease in the ion self-diffusivity with respect to the bulk, the ion conductivity remains close to that for the bulk IL, which suggests that conduction occurs in a bulk-like manner through the ions located in the pore center. In contrast, ion conduction for $f = 0.25$ and 0.5, which occurs through the IL forming a layer on the pore surface, consists of a surface mechanism that is not necessarily related to the bulk conductivity. The slowing down of the IL dynamics as f decreases indicates that the confined IL becomes “stiff”. As for the bulk IL, the dynamics of the cation is faster than that of the anion, in agreement with experiments.²² The faster dynamics of the cation arises from its homogeneous electrostatic field, which leads to a weaker interaction with its environment. In contrast, the anion, which creates a heterogeneous field with positive and negative regions, interacts strongly with its environment so that its dynamics is slower. We note that the flexible alkyl chain and planar structure of the cation backbone of the imidazolium species also contribute to its faster dynamics. Such features, which lead to the larger mobility of the cation with respect to the anion, might also explain why the decrease in the self-diffusivity upon reducing the loading f is larger for the anion than that for the cation (i.e., even when confined within the adsorbed 2D-like layer, the cation remains more mobile than the anion). The slower dynamics for the confined IL is in agreement with recent experiments that showed that the diffusion of ILs confined in hydrophilic silica mesopores is more than 10 times lower than that in the bulk.²³ Our results are also consistent with experiments showing that IL films are efficient lubricants¹⁹ with a viscosity only 1–3 orders of magnitude higher than that in the bulk.²⁰

To study the microscopic origin of the stiffening of a confined IL, ion aggregation is estimated in Figure 4a using the correlation function $P(t) = \langle \theta(t)\theta(0) \rangle$. $\theta(t) = 1$ if the cation and anion are aggregated at time t , and it is 0 otherwise. On the basis of Figure 3, cations and anions are considered “aggregated” if they are separated by a distance of <0.9 nm. $P(t)$ for the bulk and confined ILs were fitted through an exponential function and led to the aggregation times of $\tau_p \approx 3$ ns for the bulk IL and $\tau_p \approx 25$,

Table 1. Dynamical Properties of Bulk and Confined IL^a

system	$D_{\text{cation}} (10^{-8} \text{ cm}^2/\text{s})$	$D_{\text{anion}} (10^{-8} \text{ cm}^2/\text{s})$	σ (mS/cm)
$f = 0.25$	0.3	0.2	4
$f = 0.50$	4.3	3.5	7
$f = 1$	8.2	5.0	14
bulk	39.7	35.2	15

^a The maximum error is ± 0.5 on the self-diffusivities and ± 1 on the ionic conductivity.

10, and 7 ns for the confined IL with $f = 0.25$, 0.50, and 1. Larger τ_p for the confined IL are due to its slower dynamics, which makes the ions stick together over longer times. This result is consistent with experiments²⁴ in which the anion and cation dynamics in porous silica were found to be correlated over long times. The average time spent by the IL at the silica surface was estimated by calculating for the cation and anion the autocorrelation function, $R(t) = \langle \beta(t)\beta(0) \rangle$, where $\beta(t) = 1$ if the ion is located at time t within the first adsorbed layer and 0 otherwise. A cation (anion) is considered within the first adsorbed layer if it is located at a distance of >1.8 nm (1.6 nm) from the pore center. $R(t)$ decays slowly with time (Figure 4b), showing that the IL sticks to the surface because of the strong interaction between the ions and the silica atoms. At higher loadings, $R(t)$ decreases more rapidly as the gas/liquid interface and the associated free-energy barrier disappear. For a given loading, $R(t)$ for the anion decreases more slowly than that for the cation, which is consistent with the slower dynamics of the anion. This result shows that the intrinsic slower dynamics of the anion leads to a less-efficient transfer from the surface to the pore center.

Our results show that the slowing down of the dynamics and transport of the confined IL, which is associated with an increase in the correlation time, is microscopically driven by the long residence time of the IL at the surface. Decreasing f follows decreasing of the dimension of the system from a 3D-like structure (the pore with the IL) to a 2D-like structure (a single layer coating the pore surface). Such a change in the dimension of the system strongly affects the dynamics of the confined IL where Coulombic interactions have been shown to lead to large-scale collective motions.²⁵ From a practical point of view, the slowing down of the dynamics observed upon reducing the loading is qualitatively similar to that provoked upon reducing the temperature, that is, properties of the confined IL, at a given loading f and temperature T_0 , correspond to those for the bulk IL at a lower temperature T_{eff} . Figure 5 shows T_{eff} as a function of f when determined on the basis of the self-diffusivity for the cations and anions and average pairing time of the confined IL. For a given property, f can be viewed as shifting its value to that for the bulk IL taken at an effective temperature $T_{\text{eff}} \approx [1 - \exp(-nf)]T_0$, where n is an integer (Figure 5).

This study sheds light on ILs confined at the nanoscale in inorganic substrates. In agreement with recent experiments on the behavior of ILs confined in mesoporous silicas,²⁶ the magnitude of the dynamics shows that confined ILs remain liquid-like (which is confirmed by the correlation function and length). Stiffening and solid-like ordering in the direction normal to the surface are limited to the first adsorbed layers, while the single and collective bulk dynamics tend to be recovered as the IL fills the pore center. The fact that the IL loading is the key factor controlling the dynamics of confined ILs is of interest for applications such as fabricating solid electrolytes with performances approaching

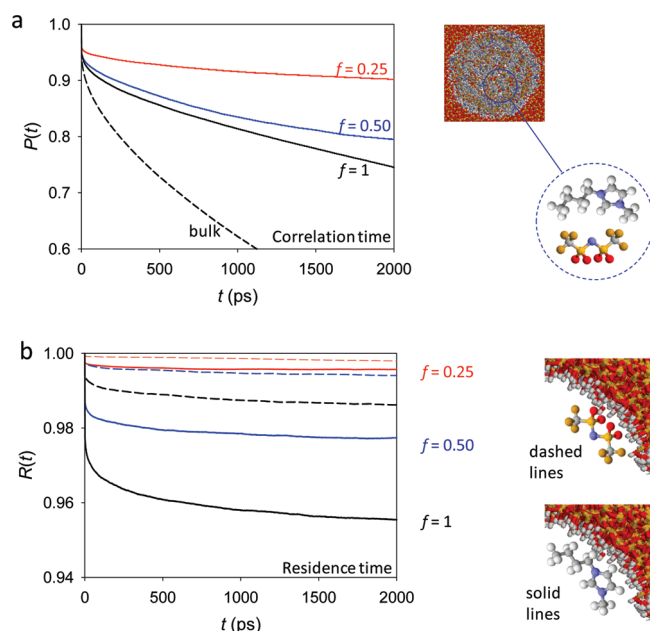


Figure 4. (a) Time correlation function $P(t)$ of anion/cation aggregation dynamics for an IL at room temperature. The solid lines are for the IL confined in the silica nanopore of diameter $D = 4.8$ nm loaded with $f = 0.25$ (red), 0.50 (blue), and 1 (black) IL. The dashed line is for the bulk. (b) Time autocorrelation function $R(t)$ of an IL in the vicinity of the surface of a silica nanopore at ambient temperature. The red, blue, and black lines are for the silica nanopore loaded with $f = 0.25$, 0.50, and 1 IL, respectively. The dashed and solid lines are for the anions and cations, respectively.

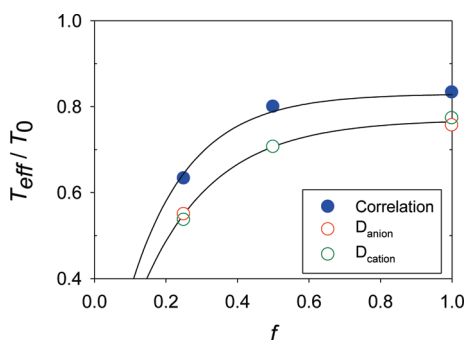


Figure 5. Effective temperature T_{eff} of a confined IL as a function of loading. T_{eff} is determined as the temperature at which the properties of the confined IL at a given loading f and temperature T_0 correspond to those for the bulk IL at a temperature T_{eff} . The different symbols correspond to T_{eff} determined on the basis of the correlation time and the cation and anion self-diffusivities. The solid lines correspond to fits of T_{eff}/T_0 by a function $[1 - \exp(-nf)]$.

those of liquids and, more generally, materials based on the immobilization of ILs in pores small enough to prevent outflow while allowing transport. The picture above, which is based on results for a typical imidazolium-based IL, is expected to be general for any ionogel or supported IL where the IL is in strong interaction with an inorganic surface (such as polar surfaces of most minerals). In contrast, hybrid materials composed of ILs having weak or other specific interactions with a solid surface may obey a different behavior and exhibit other interesting properties. In particular, transport properties can be modulated by modifying the behavior of interfacial

ILs, that is, the surface chemistry of the host material and the nature of the IL.²⁷ This opens infinite possibilities to adapt the properties of materials such as ionogels toward specific tasks.

■ ASSOCIATED CONTENT

S Supporting Information. Figure S-1, Table S-1, and procedures. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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